Vision Language Models as Explainable Classifiers for Skin Lesions

Samir Rangwalla, 2026

Background

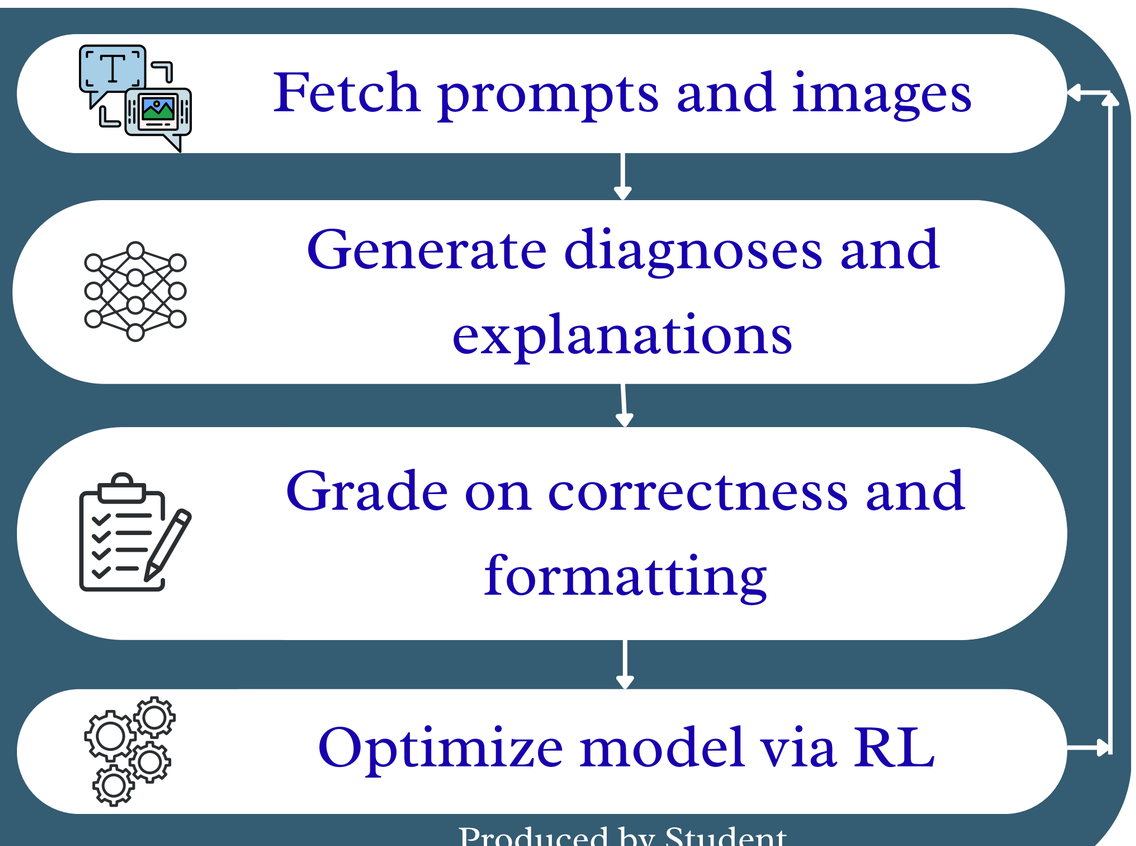
- Skin cancer affects over 20% of Americans
- AI algorithms can accurately detect skin cancer, but a lack of interpretability limits clinical adoption of AI diagnostics.
- Vision language models can generate explanations with their diagnoses
- Reinforcement learning (RL) can boost model accuracy on verifiable tasks

Objectives

- Accuracy: the model should accurately classify skin moles as malignant or benign
- Explainability: The model should generate valid, detailed explanations that justify its diagnoses

Methods

- Trained a vision–language transformer on dermoscopic images from the ISIC dataset to classify lesions as benign or malignant and generate explanations
- Generated diverse prompts by combining base templates with randomized constraints
- Used curriculum learning, ordering data from easiest to hardest based on 4-shot accuracy
- Trained using low-rank adapters
- Applied two reward functions:
  - Diagnosis correctness and explanation validity
  - Response formatting compliance



Produced by Student

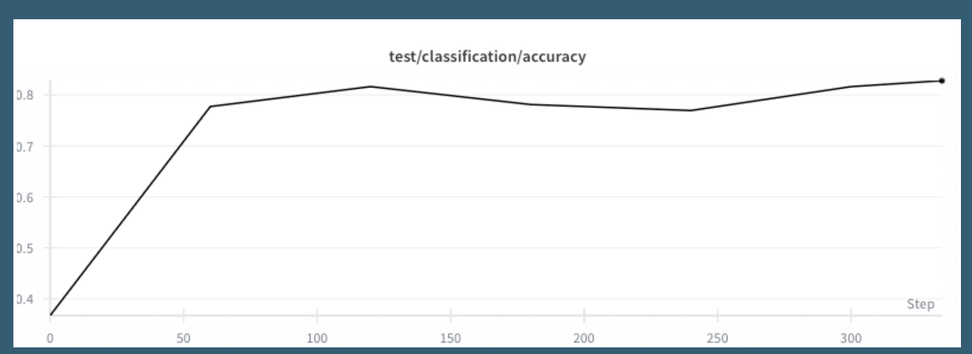
Example of Correct Diagnosis

Prompt: Patient presents with a lesion. What is your impression? Age: 55.0. Gender: male.
Model Response:
Model reasoning, summarized for brevity:

- Lesion appears as a round, well-circumscribed spot
- Lesion has a smooth border.
- The color is uniform throughout the structure, and the shape is symmetrical.
- These characteristics suggest a benign nevus.

Final output:
Diagnosis: Benign nevus.
Visual evidence: The nevus is round, symmetrical, and has a consistent shade of brown. These features are typical of a harmless mole.
Therefore, the patient should be advised on skin protection and regular monitoring for any changes in the lesion over time.

Accuracy on test set



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Accuracy rises from 36.7% to 82.8% on a held out set over the course of training.

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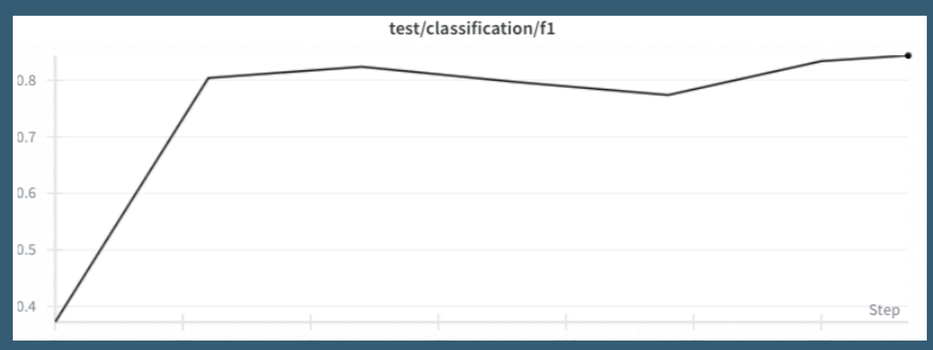
F1 Score on test set



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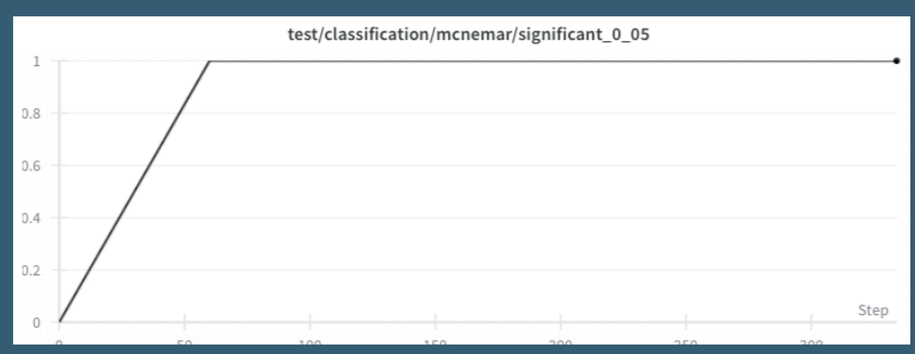
The F1 score measures how well a model balances catching real positives while avoiding false positives. It rises from 37.3% to 84.4%. Recall (true positive rate, not pictured here) rose from 34.3% to 85.0%.

McNemar Test on test set



Produced by student

The McNemar test measures if training was statistically significant or due to random variation. Statistical significance ($p<0.05$) was achieved after only 17.9% of training, and at the end of training the p-value was 1.38e-28, indicating overwhelming evidence for statistical significance.

Training Dynamics

- All charts show a trend of fast early improvement followed by a plateau
- These diminishing returns from continuing training are caused by the fact that the model learns obvious features early on, and has to learn more complex features later in training
- This behavior is common in reinforcement learning

Conclusions

- Vision Language Models can be trained with RL to accurately classify images and produce detailed explanations
- Improvements from RL are statistically significant
- The model is designed as a decision-support tool for doctors, and is not a replacement for medical judgement

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